

Soil

In most parts of the world (except for extreme deserts) the land surface is covered by soil. Most soils are mixtures of crumbling, fragmented rock and decomposing organic material. Soil contains the water that plants need and the substances that they use as nutrients. Most plants could not grow without it, and since land animals cannot survive without plants, soil is vital to life on land.

ROCK BREAKDOWN

The basis of most soil is the rock beneath it. Weathering breaks down the rock surface into smaller particles, which form the mineral content of the soil above. Soils can also develop on beds of soft sediment such as sand, and on totally organic material such as waterlogged peat that has dried out at the surface.



ORGANIC DECAY

The mineral particles in soil are mixed with decayed organic material, mostly derived from the remains of dead plants and animals. These are broken down by fungi, bacteria, and other organisms to form a dark, crumbly substance called humus. The mixture of soluble minerals and humus provides plants with the nutrients that they need to grow.



▲ CRACK AND CRUMBLE

Rainwater seeping into soluble limestone is breaking it up, forming fragments that crumble to powder. This mixes with the organic matter to form a fertile soil.